

Threat Model: Adversary Classes, Attack Complexity, and Taxonomy

Threat Model: Adversary Classes, Attack Complexity, and Taxonomy I

This section formalizes the adversary model for multiagent cognitive security. We define five adversary classes (`sec:adversary-classes`), characterize attack complexity (`sec:attack-complexity`), establish detectability metrics (`sec:detectability`), analyze adversarial capabilities (`sec:capabilities`), and present a comprehensive attack taxonomy (`sec:attack-taxonomy`).

Adversary Classes I

Definition (Adversary Class)

An adversary class Ω_k is characterized by access level, capabilities, and resource requirements.

Table 1: Adversary classification by access level and capability.

Class	Symbol	Access	Capability	Example
External	Ω_1	User input	Prompt manipulation	Jailbreak attempts
Peripheral	Ω_2	Tool/API	Data poisoning	Malicious web content
Agent-level	Ω_3	Single agent	Goal hijacking	Compromised subagent
Coordination	Ω_4	Inter-agent	Trust manipulation	MitM on messages
Systemic	Ω_5	Orchestrator	Full control	Framework compromise

`tab:adversary-classes` presents the five-tier adversary hierarchy. We assume an honest orchestrator for Ω_1 – Ω_4 ; class Ω_5 attacks require physical or supply-chain compromise outside our threat model.

Formal Mathematical Characterization of Adversary Classes I

We now provide rigorous mathematical characterizations for each adversary class Ω_k . These extend the descriptive table above with information-theoretic bounds, capability monotonicity guarantees, and formal distinguishability criteria.

Formal Mathematical Characterization of Adversary Classes II

Adversary Class Ω_1 : External Attacker

Definition (External Adversary)

An adversary $\mathcal{A} \in \Omega_1$ is characterized by the tuple:

$$\Omega_1 = \langle \mathcal{S}_{\text{input}}, \mathcal{K}_{\text{public}}, \mathcal{R}_{\text{low}}, f_{\text{detect}} \rangle \quad (1)$$

where $\mathcal{S}_{\text{input}} = \{m \in \mathcal{M} : \text{source}(m) = \text{user}\}$ is the accessible surface, $\mathcal{K}_{\text{public}}$ denotes public knowledge of the system, $\mathcal{R}_{\text{low}}$ encodes low resource requirements, and $f_{\text{detect}} : \mathcal{M} \rightarrow [0, 1]$ is the adversary's model of detection probability.

[External Attacker Bounds] For $\mathcal{A} \in \Omega_1$:

- ▶ **Access:** $|\mathcal{S}_{\text{input}}| = 1$ (single input channel)
- ▶ **Knowledge:** $H(\mathcal{K}_{\text{public}}) \leq H(\mathcal{K}_{\text{system}})$ (public knowledge bounded by system state entropy)
- ▶ **Attack complexity:** $O(1)$ — constant independent of system size
- ▶ **Detectability:** $D_{\text{min}}(\mathcal{A}_{\text{min}}) \geq D_{\text{min}}$ — highest detectability

Attack Complexity Analysis I

Definition (Resource Requirements)

Attack resources are characterized by the tuple:

$$\mathcal{R} = \langle R_C, R_K, R_A, R_P, R_{Co} \rangle \quad (12)$$

where components are defined in `tab:resource-types`.

Table 2: Attack resource taxonomy.

Resource	Symbol	Definition	Unit
Compute	R_C	Processing for attack generation	FLOPS-hours
Knowledge	R_K	System understanding required	Bits
Access	R_A	Channel availability	Interfaces
Persistence	R_P	Temporal presence required	Sessions
Coordination	R_{Co}	Multi-party synchronization	Entities

Table 3: Complexity by adversary class.

Class	R_C	R_K	R_A	R_P	R_{Co}	Complexity
Ω_1	Low	Low	1	1	1	$O(1)$
Ω_2	Medium	Medium	1–5	Variable	1	$O(\log n)$
Ω_3	High	High	1	Medium	1–2	$O(n)$
Ω_4	High	Very High	≥ 2	High	≥ 2	$O(n^2)$
Ω_5	Very High	Complete	All	Persistent	Variable	$O(2^n)$

Attack Complexity Analysis II

[Complexity Ordering]

$$\text{Complexity}(\Omega_1) < \text{Complexity}(\Omega_2) < \text{Complexity}(\Omega_3) < \text{Complexity}(\Omega_4) \\ (13)$$

Detectability Analysis I

Definition (Detectability Score)

For attack $\mathcal{A}$:

$$D_{\text{score}}(\mathcal{A}) = \alpha \cdot D_{\text{sig}} + \beta \cdot D_{\text{anom}} + \gamma \cdot D_{\text{prov}} \quad (14)$$

where $\alpha + \beta + \gamma = 1$ and components are:

- ▶ $D_{\text{sig}} \in [0, 1]$: Pattern-based detection feasibility
- ▶ $D_{\text{anom}} \in [0, 1]$: Statistical anomaly visibility
- ▶ $D_{\text{prov}} \in [0, 1]$: Causal traceability

Adversarial Capabilities I

Definition (Capability Set)

$$\mathcal{C}_{\text{adv}} = \langle C_O, C_I, C_M, C_T, C_P \rangle \quad (15)$$

with components: Observe (C_O), Inject (C_I), Modify (C_M), Timing (C_T), Persist (C_P).

Table 4: Capability matrix by adversary class.

Class	C_O	C_I	C_M	C_T	C_P
Ω_1	Input only	Direct	None	Limited	Session
Ω_2	Tool responses	API	Tool data	API timing	Tool-dep.
Ω_3	Agent state	Agent output	Beliefs	Agent timing	Memory
Ω_4	Inter-agent	Msg inject	Msg alter	Full timing	Channel
Ω_5	Complete	Complete	Complete	Complete	Complete

[Capability Monotonicity]

$$\forall i < j : \mathcal{C}_{\Omega_i} \subseteq \mathcal{C}_{\Omega_j} \quad (16)$$

Adversarial Capabilities II

[Cryptographic Limitation]

$$\forall k : \neg \text{CanBreak}(\Omega_k, \text{Crypto}) \quad (17)$$

[Byzantine Bound]

$$|\text{Compromised}| < \frac{n}{3} \quad (18)$$

[Honest Orchestrator] For adversary classes Ω_1 – Ω_4 , the orchestrator agent a_0 remains uncompromised:

$$\forall k \in \{1, 2, 3, 4\} : a_0 \notin \text{Compromised}(\Omega_k) \quad (19)$$

Attack Taxonomy I

We classify attacks into four dimensions: epistemic, behavioral, social, and temporal. `fig:threat-taxonomy` provides a visual overview of this four-dimensional classification, while `fig:comprehensive-taxonomy` presents the complete attack surface taxonomy across all five adversary classes. This formal classification is complemented by the community-maintained COGSEC ATLAS [?], which catalogs 995 cognitive security patterns across seven categories: vulnerabilities (inherent cognitive weaknesses such as in-group bias and overconfidence), exploits (methods leveraging vulnerabilities), remedies (mitigating actions), practices (established methods like Devil's Advocate and Key Assumptions Check), accelerators (factors increasing attack impact), moderators (factors influencing effect strength), and situational conditions. The Atlas employs hierarchical parent-child relationships enabling granular mapping from broad vulnerability classes to specific manifestations—a structure that aligns with our adversary class hierarchy (Ω_1 – Ω_5).

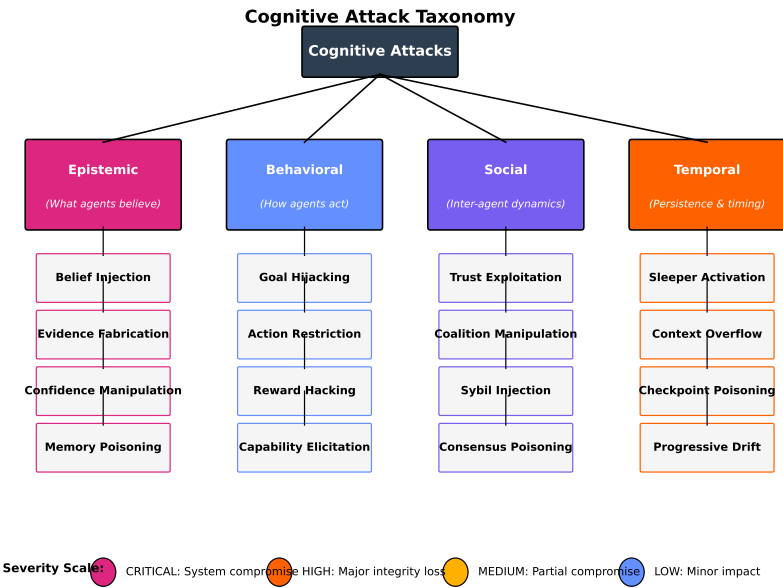


Figure 1: Four-Dimensional Threat Taxonomy: Epistemic attacks (belief